

# MATRICES

## Chapter 1: Introduction of Matrices

### 1.1 Definition 1:

A rectangular arrangement of  $mn$  numbers, in  $m$  rows and  $n$  columns and enclosed within a Bracket is called a matrix. We shall denote matrices by capital letters as A, B, C etc.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} = (a_{ij})_{m \times n}$$

A is a matrix of order  $m \times n$ .  $i^{th}$  row  $j^{th}$  column element of the matrix denoted by  $a_{ij}$

**Remark:** A matrix is not just a collection of elements but every element has assigned a definite position in a particular row and column.

### 1.2 Special Types of Matrices:

#### 1. Square matrix:

A matrix in which numbers of rows are equal to number of columns is called a square matrix.

Example:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \qquad B = \begin{pmatrix} 2 & 5 & -8 \\ 0 & -3 & -4 \\ 6 & 8 & 9 \end{pmatrix}$$

#### 2. Diagonal matrix:

A square matrix  $A = (a_{ij})_{n \times n}$  is called a diagonal matrix if each of its non-diagonal element is zero.

That is  $a_{ij} = 0$  if  $i \neq j$  and at least one element  $a_{ii} \neq 0$ .

Example:

$$A = \begin{pmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{pmatrix}$$

$$B = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 9 \end{pmatrix}$$

### 3. Identity Matrix

A diagonal matrix whose diagonal elements are equal to 1 is called identity

Matrix and denoted by  $I_n$ .

$$\text{That is } a_{ij} = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

Example:

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

### 4. Upper Triangular matrix:

A square matrix said to be a Upper triangular matrix if  $a_{ij} = 0$  if  $i > j$ .

Example:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{pmatrix} \quad B = \begin{pmatrix} 2 & 0 & 8 \\ 0 & -2 & 5 \\ 0 & 0 & 7 \end{pmatrix}$$

### 5. Lower Triangular Matrix:

A square matrix said to be a Lower triangular matrix if  $a_{ij} = 0$  if  $i < j$ .

Example:

$$A = \begin{pmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \quad B = \begin{pmatrix} -1 & 0 & 0 \\ 7 & 0 & 0 \\ 9 & 6 & 2 \end{pmatrix}$$

### 6. Symmetric Matrix:

A square matrix  $A = (a_{ij})_{n \times n}$  said to be a symmetric if  $a_{ij} = a_{ji}$  for all  $i$  and  $j$ .

Example:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{pmatrix} \quad B = \begin{pmatrix} 8 & -2 & 7 \\ -2 & -9 & 3 \\ 7 & 3 & 5 \end{pmatrix}$$

### 7. Skew- Symmetric Matrix:

A square matrix  $A = (a_{ij})_{n \times n}$  said to be a skew-symmetric if  $a_{ij} = -a_{ji}$  for all i and j.

Example:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ -a_{12} & a_{22} & a_{23} \\ -a_{13} & -a_{23} & a_{33} \end{pmatrix} \quad B = \begin{pmatrix} 8 & -2 & 7 \\ 2 & -9 & 3 \\ -7 & -3 & 5 \end{pmatrix}$$

### 8. Zero Matrix:

A matrix whose all elements are zero is called as Zero Matrix and order  $n \times m$  Zero.

Matrix denoted by  $0_{n \times m}$ .

Example:

$$0_{3 \times 2} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

### 9. Row Vector

A matrix consists a single row is called as a row vector or row matrix.

Example:

$$A = (a_{11} \quad a_{12} \quad a_{13}) \quad B = (7 \quad 4 \quad -3)$$

### 10. Column Vector

A matrix consists a single column is called a column vector or column matrix.

Example:

$$A = \begin{pmatrix} a_{11} \\ a_{21} \\ a_{31} \end{pmatrix} \quad B = \begin{pmatrix} 9 \\ -7 \\ 3 \end{pmatrix}$$

## Chapter 2: Matrix Algebra

### 2.1 Equality of two matrices:

Two matrices A and B are said to be equal if

- (i) They are same order
- (ii) Their corresponding elements are equal.

That is if  $A = (a_{ij})_{m \times n}$  and  $B = (b_{ij})_{m \times n}$  then  $a_{ij} = b_{ij}$  for all i and j.

### 2.2. Scalar multiple of a matrix

Let k be a scalar then scalar product of matrix  $A = (a_{ij})_{m \times n}$  given denoted by kA and

given by  $kA = (ka_{ij})_{m \times n}$  or

$$kA = \begin{pmatrix} ka_{11} & ka_{12} & \cdots & ka_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ ka_{m1} & ka_{m2} & \cdots & ka_{mn} \end{pmatrix}$$

### 2.3. Addition of two matrices:

Let  $A = (a_{ij})_{m \times n}$  and  $B = (b_{ij})_{m \times n}$  are two matrices with same order then sum of the two matrices are given by

$$A + B = (a_{ij})_{m \times n} + (b_{ij})_{m \times n} = (a_{ij} + b_{ij})_{m \times n}$$

**Example 2.1:** let

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 4 & 5 & -4 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & 0 & 2 \\ -1 & 1 & 8 \end{pmatrix}$$

Find (i) 5B (ii) A + B (iii) 4A – 2B (iv) 0 A

### 2.4. Multiplication of two matrices:

Two matrices A and B are said to be confirmable for product AB if number of columns in

A equals to the number of rows in matrix B. Let  $A = (a_{ij})_{m \times n}$  and  $B = (b_{ij})_{n \times r}$  be two

Matrices the product matrix  $C = AB$ , is matrix of order  $m \times r$  where

$$c_{ij} = \sum_{k=1}^n a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots \cdots \cdots + a_{in}b_{nj}$$

Example 2.2: Let  $A = \begin{pmatrix} 1 & -2 & 3 & 4 \\ 4 & 5 & -1 & 1 \end{pmatrix}$  and  $B = \begin{pmatrix} 2 & 3 \\ -5 & 0 \\ 6 & -2 \\ -1 & -3 \end{pmatrix}$

Calculate (i) AB (ii) BA (iii) is AB = BA ?

## 2.5. Integral power of Matrices:

Let A be a square matrix of order n, and m be positive integer then we define

$$A^m = A \times A \times A \cdots \cdots \cdots \times A \text{ (m times multiplication)}$$

## 2.6. Properties of the Matrices:

Let A, B and C are three matrices and  $\lambda$  and  $\mu$  are scalars then

- (i)  $A + (B + C) = (A + B) + C$  Associative Law
- (ii)  $\lambda (A + B) = \lambda A + \lambda B$  Distributive law
- (iii)  $\lambda (\mu A) = (\lambda \mu) A$  Associative Law
- (iv)  $(\lambda A)B = \lambda(AB)$  Associative Law
- (v)  $A(BC) = (AB)C$  Associative Law
- (vi)  $A(B + C) = AB + AC$  Distributive law

## 2.7 Transpose

The transpose of matrix  $A = (a_{ij})_{m \times n}$  written  $A^t$  (*A or A<sup>T</sup>*) is the matrix obtained by writing the rows of A in order as columns.

$$\text{That is } A^t = (a_{ji})_{n \times m}$$

### Properties of Transpose

- i.  $(A + B)^t = (A^t + B^t)$
- ii.  $(A^t)^t = A$
- iii.  $(kA)^t = kA^t$  For scalar k.
- iv.  $(AB)^t = B^t A^t$

Example 2.3: Using the following matrices A and B, Verify the transpose properties

$$A = \begin{pmatrix} 1 & -1 & 2 \\ 5 & -4 & 3 \\ 1 & -2 & -3 \end{pmatrix} \quad B = \begin{pmatrix} -2 & 6 & -2 \\ -1 & 0 & 1 \\ -2 & 1 & 0 \end{pmatrix}$$

**2.8 A square matrix A is said to be symmetric if  $A = A^t$ .**

Example:

$$A = \begin{pmatrix} 1 & -1 & 1 \\ 5 & -4 & -2 \\ 1 & -2 & -3 \end{pmatrix}, \text{ A is symmetric by the definition of symmetric matrix. Then}$$

$$A^t = \begin{pmatrix} 1 & -1 & 1 \\ 5 & -4 & -2 \\ 1 & -2 & -3 \end{pmatrix} \quad \text{That is } A = A^t.$$

**2.9 A square matrix A is said to be skew- symmetric if  $A = -A^t$ .**

Example:

$$A = \begin{pmatrix} 1 & 3 & -1 \\ -3 & -5 & 8 \\ 1 & 8 & 9 \end{pmatrix}$$

- (i)  $AA^t$  and  $A^tA$  are both symmetric
- (ii)  $A + A^t$  is a symmetric matrix
- (iii)  $A - A^t$  is a skew-symmetric matrix
- (iv) If A is a asymmetric matrix and m is any positive integer then  $A^m$  is also symmetric
- (v) If A is skew-symmetric matrix then odd integral power of A is skew-symmetric, while positive even integral powers of A are symmetric.

If A and B are symmetric metrics then

- (vi)  $(AB + BA)$  is symmetric
- (vii)  $(AB - BA)$  is skew-symmetric

**Exercise 2.1:**

Verify the (i) ,(ii) and (iii) using the following matrix,  $A = \begin{pmatrix} 1 & 3 & -5 \\ -3 & -5 & 10 \\ 1 & 8 & 9 \end{pmatrix}$

## Chapter 3: Determinant, Minor and Adjoint Matrices

### Definition 3.1:

Let  $A = (a_{ij})_{n \times n}$  be a square matrix of order n, then the number  $|A|$  called determinant of the matrix A.

(i) Determinant of  $2 \times 2$  matrix

$$\text{Let } A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \text{ then } |A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

(ii) Determinant of  $3 \times 3$  matrix

$$\text{Let } B = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\text{Then } |B| = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$|B| = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{31}a_{22})$$

Exercise 3.1: Calculate the determinants of the following matrices

$$(i) \ A = \begin{pmatrix} 1 & 3 & 4 \\ 2 & 6 & 8 \\ 1 & 9 & 5 \end{pmatrix} \quad B = \begin{pmatrix} 2 & -3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 1 \end{pmatrix}$$

### 3.1 Properties of the Determinant:

a) The determinant of a matrix A and its transpose are equal.

$$|A| = |A^t|$$

b) Let A be a square matrix

(i) If A has a row (column) of zeros then  $|A| = 0$

(ii) If A has two identical rows ( or columns) then  $|A| = 0$

c) If A is triangular matrix then  $|A|$  is product of the diagonal elements.

d) If A is a square matrix of order n and k is a scalar then  $|kA| = k^n |A|$

### 3.2 Singular Matrix

If  $A$  is square matrix of order  $n$ , the  $A$  is called singular matrix when and non-singular otherwise.

### 3.3. Minor and Cofactors:

Let  $A = (a_{ij})_{n \times n}$  is a square matrix. Then  $M_{ij}$  denote a sub matrix of  $A$  with order  $(n - 1) \times (n - 1)$  obtained by deleting its  $i^{th}$  row and  $j^{th}$  column. The determinant  $|M_{ij}|$  is called the minor of the element  $a_{ij}$  of  $A$ .

The cofactor of  $a_{ij}$  denoted by  $A_{ij}$  and is equal to  $-1^{i+j}|M_{ij}|$

**Exercise 3.2:** Let  $A = \begin{pmatrix} 5 & 4 & 2 \\ 2 & 3 & 1 \\ 3 & -2 & -1 \end{pmatrix}$

(i) Compute determinant of  $A$ .

(ii) Find the cofactor matrix.

### 3.4. Adjoin Matrix:

The transpose of the matrix of cofactors of the element  $a_{ij}$  of  $A$  denoted by  $adj A$  is called adjoint of matrix  $A$ .

**Example 3.3:** Find the adjoint matrix of the above example.

### Theorem 3.1:

For any square matrix  $A$ ,

$A(adj A) = (adj A)A = |A|I$  where  $I$  is the identity matrix of same order.

**Proof:** Let  $A = (a_{ij})_{n \times n}$

Since  $A$  is a square matrix of order  $n$ , then also in same order.

Consider

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & \cdots & a_{nn} \end{pmatrix} \text{ then}$$

$$\text{adj}A = \begin{pmatrix} A_{11} & A_{12} & \cdots & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & \cdots & A_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ A_{n1} & A_{n2} & \cdots & \cdots & A_{nn} \end{pmatrix}$$

Now consider the product  $A(\text{adj}A)$

$$\begin{aligned} A(\text{adj}A) &= \begin{pmatrix} a_{11} & a_{12} & \cdots & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} & \cdots & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & \cdots & A_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ A_{n1} & A_{n2} & \cdots & \cdots & A_{nn} \end{pmatrix} \\ &= \begin{pmatrix} \sum_{j=1}^n a_{1j}A_{1j} & \sum_{j=1}^n a_{1j}A_{2j} & \cdots & \cdots & \sum_{j=1}^n a_{1j}A_{nj} \\ \sum_{j=1}^n a_{2j}A_{1j} & \sum_{j=1}^n a_{2j}A_{2j} & \cdots & \cdots & \sum_{j=1}^n a_{2j}A_{nj} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \sum_{j=1}^n a_{nj}A_{1j} & \sum_{j=1}^n a_{nj}A_{2j} & \cdots & \cdots & \sum_{j=1}^n a_{nj}A_{nj} \end{pmatrix} \\ &= \begin{pmatrix} |A| & 0 & \cdots & 0 \\ 0 & |A| & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & |A| \end{pmatrix} \end{aligned}$$

(as we know that  $\sum_{j=1}^n a_{ij}A_{ij} = |A|$  and  $\sum_{j=1}^n a_{ij}A_{kj} = 0$  when  $i \neq k$ )

$$= |A| \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

$$= |A|I_n \text{ Where } I_n \text{ is unit matrix of order } n$$

**Theorem 3.2:** If  $A$  is a non-singular matrix of order  $n$ , then  $|adj A| = |A|^{n-1}$

**Proof:** By the theorem 1

$$A(adj A) = |A|I$$

$$|A(adj A)| = ||A|I|$$

$$|A||adj A| = |A|^n$$

$$|adj A| = |A|^{n-1}$$

Theorem 3.3: If  $A$  and  $B$  are two square matrices of order  $n$  then

$$adj(AB) = (adj B)(adj A)$$

**Some results of adjoint**

- (i) For any square matrix  $A$   $(adj A)^t = adj A^t$
- (ii) The adjoint of an identity matrix is the identity matrix.
- (iii) The adjoint of a symmetric matrix is a symmetric matrix.

## Chapter 4: Inverse of a Matrix and Elementary Row Operations

### 4.1 Inverse of a Matrix

**Definition 4.1:**

If  $A$  and  $B$  are two matrices such that  $AB = BA = I$ , then each is said to be inverse of the other. The inverse of  $A$  is denoted by  $A^{-1}$ .

**Theorem 4.1: (Existence of the Inverse)**

The necessary and sufficient condition for a square matrix  $A$  to have an inverse is that  $|A| \neq 0$  (That is  $A$  is non-singular).

**Proof:**

(i) The necessary condition

Let A be a square matrix of order n and B is inverse of it, then

$$AB = I$$

$$|AB| = |A||B| = |I|$$

$$\text{Therefore } |A| \neq 0$$

(ii) The sufficient condition

If  $|A| \neq 0$ , then we define the matrix B such that

$$B = \frac{1}{|A|} (\text{adj } A)$$

$$\text{Then } AB = A \frac{1}{|A|} (\text{adj } A) = \frac{1}{|A|} A(\text{adj } A)$$

$$= \frac{1}{|A|} |A| I = I$$

$$\text{Similarly } BA = \frac{1}{|A|} (\text{adj } A) A = \frac{1}{|A|} A(\text{adj } A) = \frac{1}{|A|} |A| I = I$$

Thus  $AB = BA = I$  Hence B is inverse of A and is given by  $A^{-1} = \frac{1}{|A|} (\text{adj } A)$

**Theorem 4.2: (Uniqueness of the Inverse)**

Inverse of a matrix if it exists is unique.

**Proof:** Let B and C are inverse s of the matrix A then

$$AB = BA = I \text{ and } AC = CA = I$$

$$B(AC) = BI$$

$$(BA)C = B$$

$$C = B$$

Example 6: Let  $A = \begin{pmatrix} 2 & 3 & -4 \\ 0 & -4 & 2 \\ 1 & -1 & 5 \end{pmatrix}$  find  $A^{-1}$

**Theorem 4.3: (Reversal law of the inverse of product)**

If A and B are two non-singular matrices of order n, then  $(AB)$  is also non-singular and

$$(AB)^{-1} = B^{-1}A^{-1}$$

Proof:

Since A and B are non-singular  $|A| \neq 0$  and  $|B| \neq 0$  therefore  $|A||B| \neq 0$ , then  $|AB| \neq 0$ .

Consider  $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1}$

$$= AIA^{-1}AA^{-1} = I \quad \dots\dots\dots (1)$$

Similarly  $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B$

$$= B^{-1}IB = B^{-1}B = I \quad \dots\dots\dots (2)$$

From (1) and (2)

$$(AB)(B^{-1}A^{-1}) = (B^{-1}A^{-1})(AB) = I$$

Therefore by the definition and uniqueness of the inverse  $(AB)^{-1} = B^{-1}A^{-1}$

**Corollary4.1:** If  $A_1A_2 \dots \dots \dots A_m$  are non-singular matrices of order n, then

$$(A_1A_2 \dots \dots \dots A_m)^{-1} = A_1^{-1}A_2^{-1} \dots \dots \dots A_m^{-1}$$

**Theorem 4.4:** If A is a non-singular matrix of order n then  $(A^t)^{-1} = (A^{-1})^t$

**Theorem 4.5:** If A is a non-singular matrix, k is non zero scalar, then  $(kA)^{-1} = \frac{1}{k}A^{-1}$

**Theorem 4.6:** If A is a non-singular matrix then

$$|A^{-1}| = \frac{1}{|A|}$$